EXECUTIVE SUMMARY
GLIDECART

JANUARY 2026

GlideCart Quantum AI Grocery Navigation

GLIDECART is an optimized quantum framework for autonomous vehicle that navigate while it shops and delivers to and from grocery stores. Thereby, it requires a robust error correction. Futhermore, Glidecart is my project that I work on while in graduate school at Cybersecurity Management in Purdue Global. Consequently, the project draws on Professor Lukin's research in quantum optics and scalable quantum processors (Bluvstein et al., 2024). Thereby, I can reconfigure atom arrays for logical qubits.

The core of GlideCart starts with quantum principles for grocery store efficiency. Thus, variational quantum is called eigensolver simulations, and GlideCart's autonomous vehicle module. Futhermore, in 2025 they reduce navigation from 0.625 to 0.5 minutes. Futhermore, Glidecart aligns with Lukin's high-fidelity entangled gates as multi-qubit operations (Evered et al., 2023). Consequently, integrate scalable improvements for quantum apps in grocery shopping. Specifically, they minimize congestion at peak hours in grocery environments.

Futhermore, GlideCart has quantum error correction techniques and secure data in a networked grocery systems. Next. it mitigates cyber threats and remain resilient during interference. For example: capabilities extend atom-based quantum simulators to hybrid classical-quantum ecosystems. As a result, it transforms from theoretical idea into an everday quantum autonomous vehicle.

The Harvard Quantum Initiative offers an opportunity to advance my research. Conversely, architecture of GlideCart aligns HQI's focus with quantum networks and logical qubit scalability. Hence, I can collaborate on atom-array dynamic processors. In the end, collaboration with GlideCart and HQI's is crucial to successfully explore fault-tolerant quantum systems in groceries. Ultimately, it can deliver impact through quantum grocery experiences.